

Bo-Yu Chen 陳柏昱

Researcher and builder working across music AI, creative tools, musical interfaces, and human-computer interaction.

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RESEARCH INTERESTS

Human-Computer Interaction (HCI) · Music Information Retrieval (MIR) · Computational Media · AI Agents · Creative Support Tools · Musical Interfaces

EDUCATION

National Taiwan University (NTU)
M.S. in Networking and Multimedia (Dept. of CSIE)

- Master's Thesis:** *FlueBricks: A Modular Toolkit for Interactive Flute-like Instrument Systems* [C1] (Advisor: Prof. Lung-Pan Cheng)
- Thesis Honors:** Winner of **TAICHI 2026 Master's Thesis Honorable Mention Award** Top 16.6%

Sep 2021 – Aug 2025
Taipei, Taiwan

National Taiwan University of Science and Technology (NTUST)
B.S. in Electrical Engineering

Sep 2013 – Jun 2018
Taipei, Taiwan

RESEARCH EXPERIENCE

Computational Physicality Lab, National Taiwan University
Graduate Research Assistant (Advisor: Prof. Lung-Pan Cheng)

- Designed FlueBricks**, a modular physical-computing toolkit enabling acoustic replication and customizable design of flute-like instruments as Master's thesis research. [C1]
- Engineered parametric CAD assemblies in Fusion 360** for digital fabrication, and conducted user evaluations across musician skill levels to validate expressive playability.

Sep 2021 – Aug 2025
Taipei, Taiwan

Creative AI Lab, Sony Group Corporation
Research Intern (Advisors: Dr. Yuki Mitsufuji, Dr. Marco A. Martínez Ramírez, Wei-Hsiang Liao)

- Developed DJTransGAN**, the first generative model for automated cross-track DJ transitions using DDSP and GANs. [C2]
- Open-sourced DJtransGAN**  and its **data-generation pipeline** , earning **120+ GitHub stars** and community interest.

Nov 2020 – Sep 2021
Tokyo, Japan (Remote)

Music and AI Lab, Academia Sinica
Research Assistant (Advisor: Prof. Yi-Hsuan Yang)

- Pioneered loop-based music AI research** spanning compatibility assessment, generation, and benchmarking. [C3] [C4] [C5]
- Mentored 5+ junior researchers** and **co-authored 5 papers** on music AI.

Jul 2019 – Sep 2023
Taipei, Taiwan

WORK EXPERIENCE

Rhythm Culture Corporation 
Technical Lead

- Architecting Rytho**, a local-first creative workspace that enables AI agents to listen to, understand, search for, and create music alongside people.
- Led R&D on LLM-driven music agents**, developing conversational feedback agents for music production and reasoning systems for chord analysis. [W1] [W2]
- Managed engineering teams for government R&D projects** (TAICCA, Taipei City) and **TMC-CL1** , a web music interface commissioned by Taipei Music Center.

Jul 2023 – Present
Taipei, Taiwan

Zone Sound Creative Corporation
Interactive Engineer

- Architected interactive web audio interfaces and networked performance systems**, developing a vocal sampler for Vocal Asia (Japan) [A2] and a p5.js/Node.js WebSocket engine bridging mobile audiences to Max/MSP stage synthesis [A4].
- Developed a web demo showcasing Taiwanese electronic music** in collaboration with electronic music artists for MIDEM (Cannes, France). [A3]

Mar 2019 – Aug 2019
Taipei, Taiwan

TEACHING EXPERIENCE

BLND Creative Coding School

Aug 2020 – Jul 2023

Instructor

Taipei, Taiwan

- Taught generative AI art (GANs, Stable Diffusion workflows) and physical computing with Arduino to creative practitioners.

OrangeApple Children Programming School

Jul 2015 – Sep 2019

Instructor

Taipei, Taiwan

- Taught algorithms and C++/JS to K-12 students; coached high school students for the APCS examination.

PUBLICATIONS

C = Conference papers · W = Workshop papers

C1 FlueBricks: A Construction Kit of Flute-like Instrument for Acoustic Reasoning

Bo-Yu Chen, Chiao-Wei Huang, Lung-Pan Cheng

ACM Conference on Human Factors in Computing Systems (CHI 2026)

Acceptance ~25%

TAICHI 2025 · Best Paper



W1 Enhancing Automatic Chord Recognition through LLM Chain-of-Thought Reasoning

Chih-Cheng Chang, Bo-Yu Chen, Lu-Rong Chen, Li Su

International Society for Music Information Retrieval Conference (ISMIR 2025) – LLM4Music Workshop 

W2 AI TrackMate: Finally, Someone Who Will Give Your Music More Than Just "Sounds Great!"

Yi-Lin Jiang, Chia-Ho Hsiung, Yen-Tung Yeh, Lu-Rong Chen, Bo-Yu Chen

Conference on Neural Information Processing Systems (NeurIPS 2024) – Creative AI Track

Acceptance 31%



C2 Automatic DJ Transitions with Differentiable Audio Effects and Generative Adversarial Networks

Bo-Yu Chen, Wei-Han Hsu, Wei-Hsiang Liao, Marco A. Martínez Ramírez, Yuki Mitsufuji, Yi-Hsuan Yang

IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP 2022)

Acceptance 47%



C3 Exploiting Pre-trained Feature Networks for Generative Adversarial Networks in Audio-Domain Loop Generation

Yen-Tung Yeh, Bo-Yu Chen, Yi-Hsuan Yang

International Society for Music Information Retrieval Conference (ISMIR 2022)

Acceptance 43%



C4 A Benchmarking Initiative for Audio-Domain Music Generation Using the Freesound Loop Dataset

Tun-Min Hung, Bo-Yu Chen, Yen-Tung Yeh, Yi-Hsuan Yang

International Society for Music Information Retrieval Conference (ISMIR 2021)

Acceptance 34%



C5 Neural Loop Combiner: Neural Network Models for Assessing the Compatibility of Loops

Bo-Yu Chen, Jordan Smith, Yi-Hsuan Yang

International Society for Music Information Retrieval Conference (ISMIR 2020)

Acceptance 38%



INTERACTIVE ART & INSTALLATIONS

(Works created in collaboration with artists, musicians, and designers)

A1 Iridescent Vision

2019

A web-based interactive music video and light installation exploring human civilization and imagined future technology.

With DJ Sonia Calico and visual artist Celso Urroz · Presented at FINAL Club, Taipei, Taiwan

A2 Taptap

2019

A web-based interface that lets visitors play music with a cappella samples.

With O-KAI SINGERS · Presented at Vocal Asia Festival, Kagawa, Japan

A3	FusionTaiwanBeat 	2019
A web demo showcasing Taiwanese electronic music. <i>With DJ Sonia Calico and visual artist Celso Urroz · Presented at MIDEM, Cannes, France</i>		
A4	Duplex Transmission 	2019
A web-based system that connects audiences and performers during live performance. <i>With new media artist Po-Hao Chi · Presented at Songshan Cultural Park, Taipei, Taiwan</i>		
A5	Mimo No.Tent 	2019
A virtual planetarium that brings a starry sky inside a tent. <i>With designers Yi-Yun Liao and Jing-Ting Huang · Presented at TaiwanTech Design Grad Show, Taipei, Taiwan</i>		
A6	YuYu 	2018
An AR service that delivers customized messages through Chinese pastries. <i>With designer Ya-Cin Siao · Presented at TaiwanTech Design Grad Show, Taipei, Taiwan</i>		
A7	Message Container 	2017
A breath-driven installation that responds with haptic, sound, and visual feedback. <i>With new media artist Wen-Fei Zhang · Presented at TaiwanTech Design Grad Show & Venue, Taipei, Taiwan</i>		

OPEN-SOURCE SOFTWARE & TOOLKITS

OpenMIRLab 	Nov 2025 – Present
Founder & Lead Maintainer	Open-Source
• Building OpenMIRLab to make music AI research easier to use and build on, bringing 30+ MIR models into a consistent, easy-to-install toolkit. • Standardized unified inference APIs and checkpoint registries, stripping heavy training dependencies into lightweight, pip-installable packages with verified model outputs.	

HONORS & AWARDS

Master's Thesis Honorable Mention Award · TAICHI Thesis Award for <i>FlueBricks</i> (published as [C1])	Top 16.6%	2026
Best Paper Award  · Taiwan Computer-Human Interaction Society (TAICHI) for <i>FlueBricks</i> [C1]		2025
Overseas Visit Delegate · Epoch Foundation Young Entrepreneurs of the Future	Top 9% · 16/175	2017
3rd Prize · Ministry of Economic Affairs (MOEA) GetFresh Creativity Competition		2017

ACADEMIC SERVICE & REVIEWING

International Society for Music Information Retrieval Conference (ISMIR)	2020 – 2026
IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)	2025
Conference on Neural Information Processing Systems (NeurIPS) – Creative AI Track	2025

EXTRACURRICULAR ACTIVITIES

Musical Theater: Rose of the Endless Night  	Feb 2015 – Jun 2015
Director, Music Theater Course Final Public Performance (NTUST)	Taipei, Taiwan
• Authored a stage adaptation of a Taiwanese musical and directed 60+ cast and crew members for a live stage production with 400+ audience.	
NTUST Men's Football Team	Jul 2016 – Jul 2017
Vice Captain, Position: Defensive Midfielder	Taipei, Taiwan
• Coordinated squad training and match operations across 30 collegiate athletes and coaching staff. • Led team to 4th Place in the National Inter-University Football Tournament and 5th Place in the National Futsal Tournament.	